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REGULAR PAPER

Smile: enhancing low-light images in lightweight networks via exposure-aware non-reference losses

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Abstract
Low-light image enhancement (LLIE) is a critical task in computer vision, with applications ranging from autonomous driving to medical imaging. Existing LLIE methods often struggle to balance enhancing underexposed regions while preserving well-lit areas, especially in lightweight networks. To address this challenge, we propose SMILE, a non-reference framework designed for lightweight, skip-free architectures. SMILE employs a monotonic increasing function (MIF) to ensure consistent brightening and stable training. Two novel non-reference loss functions—Underexposure Correction Loss (UCL) and Exposure Preservation Loss (EPL)—are introduced to provide patch-wise, exposure-aware supervision. These components jointly enhance dark regions while maintaining the natural fidelity of well-exposed areas. With only 3.3k parameters and a 77% reduction in inference time, SMILE achieves superior perceptual quality compared to state-of-the-art methods. Extensive experiments and ablation studies demonstrate that UCL and EPL significantly improve training stability and enable localized exposure control. Our implementation will be made publicly available at https://doi.org/10.5281/zenodo.15624362.

Keywords Low-light image enhancement (LLIE) · Exposure-aware loss functions · Lightweight neural networks · Non-reference Learning · Monotonic modulation

1 Introduction

Low-light image enhancement (LLIE) is a fundamental task in computer vision, supporting safety-critical applications such as autonomous driving, surveillance, and medical imaging [1–3]. These systems increasingly require solutions that not only improve perceptual quality but also meet strict constraints on model size and inference speed—particularly when deployed on edge devices.

Recent LLIE approaches based on curve estimation [4, 5] have shown promising results by learning pixel-wise intensity mappings. However, these methods often employ unconstrained modulation functions that allow both brightening and darkening, which can introduce instability, especially in non-reference or lightweight architectures. Additionally, most loss functions used in LLIE [4, 5] lack spatial awareness and apply global supervision uniformly across all regions of an image. This can result in over-enhancement, color distortion, and poor generalization to diverse lighting conditions.

We address these challenges with SMILE (Simplified Monotonic Increasing Learning for Enhancement), a

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non-reference LLIE framework that enables stable, high-quality enhancement using a compact, skip-free architecture. SMILE is built upon a monotonic increasing function (MIF) with two novel exposure-aware loss functions. These loss functions provide localized, patch-level supervision that selectively enhances underexposed areas while preserving details in well-lit regions.

In summary, this research makes the following contributions:

- We introduce two novel exposure-aware loss functions—Underexposure Correction Loss (UCL) and Exposure Preservation Loss (EPL)—which enable spatially targeted supervision without reference images, guiding enhancement only where needed.
- We develop a modulation strategy based on a Monotonic Increasing Function (MIF) that enforces consistent brightening behavior across all luminance levels, eliminating the need for skip connections or deep architectures.
- We propose SMILE, a non-reference LLIE framework with only 3.3k parameters that outperforms SOTA methods in both perceptual quality and computational efficiency, demonstrating that well-aligned supervision can replace architectural complexity.

SMILE achieves SOTA performance while operating with minimal architectural complexity, making it well-suited for real-time and low-power applications. We validate its effectiveness through extensive experiments and ablation studies.

2 Related work

2.1 Supervision strategies in LLIE

Loss functions play a central role in guiding LLIE models during training. Early approaches relied heavily on reference-based losses, such as pixel-wise MSE or perceptual loss [6], which require ground-truth paired images to preserve structural fidelity and visual quality [7–9]. Generative methods later introduced adversarial and contrastive losses [9, 10] to improve realism, but often suffered from unstable training and hallucinated details.

More recent non-reference losses [11–13] eliminate the need for paired supervision by leveraging handcrafted priors or no-reference image quality metrics (e.g., BRISQUE, NIQE). These methods aim to provide non-reference setups, as seen in approaches like SCI [14]. However, most of these losses apply uniform penalties across the image, failing to account for spatial variations in exposure. This global supervision can result in excessive brightening or distortion, especially in already well-lit regions.

Recent work in other visual domains has also demonstrated the benefits of task-specific and perceptually grounded decision strategies. For example, deep feature retrieval in portrait photography [15] and deep fusion models for 3D facial attractiveness prediction [16] rely on localized feature cues to guide output decisions. These approaches reflect a growing recognition that spatially adaptive, content-aware guidance can improve visual quality—an idea we explore from a new perspective in the context of LLIE.

In contrast, our work introduces exposure-aware loss functions that adapt supervision based on local luminance, targeting only underexposed areas for enhancement while preserving well-lit regions.

2.2 Curve-based modulation for enhancement

Curve-based models [4, 17] have gained popularity in LLIE due to their compactness and interpretability. These methods enhance images using nonlinear functions that adjust pixel intensities based on parametric modulation. Notable examples include Zero-DCE and its variants [18–21], which estimate pixel-wise enhancement curves in a fully non-reference manner.

Despite their success, these models often lack functional constraints, allowing modulation curves to decrease brightness or oscillate in mid-tones. Such non-monotonic behavior contradicts the LLIE objective and can destabilize training, particularly in lightweight architectures. Recent efforts to improve stability (e.g., LEES-Net) [4] introduce complex modules like attention or multi-branch designs, increasing model size and inference cost.

Our approach addresses this instability by enforcing monotonicity through a dedicated modulation function, allowing effective and stable enhancement even in skip-free networks.

2.3 Model architectures for efficient LLIE

Many high-performing LLIE models rely on heavy architectures to compensate for weak supervision and unconstrained enhancement behavior. Retinexformer [22] and SNR-Net [23] leverage transformers or deep feature extractors to model global illumination. Other works [24–26] incorporate residual blocks, attention, or multi-path structures to improve quality—but often at the expense of speed and portability.

In contrast, lightweight models like LiCENt [27] or LVENet [28] attempt to reduce complexity through decomposed processing or specialized layers. LPNet [29] applied a coarse-to-fine strategy with luminance-aware refinement

to efficiently reconstruct normal-light images. Additionally, transformer-based strategies [30–32] have been adapted for lightweight scenarios; for example, STAR [33] proposed a lightweight structure-aware transformer for real-time image enhancement without relying on large transformer backbones. However, despite being lightweight, many of these models still rely on specially designed components to preserve enhancement quality.

Outside the domain of LLIE, architectural innovations in 3D vision such as TMSDNet [34], which incorporates multi-scale and dense attention structures, and RADepthNet [35], which integrates reflectance and boundary guidance, support the broader principle that task-specific structural design can improve performance without relying on excessive network depth. In a related context, video super-resolution methods like FFFN [36] adopt frame-wise feedback refinement to enhance quality with minimal overhead. In a related image restoration domain, blind super-resolution research has also explored alternative ways to improve robustness and efficiency beyond purely increasing network depth. For example, degradation-aware and contrastive learning strategies have been used to learn representations that are resilient to unknown and spatially varying degradations [37, 38]. Separately, lightweight super-resolution models such as CATANet demonstrate that careful architectural design can substantially reduce parameters and computational cost while maintaining strong reconstruction performance [39]. Although these methods address different restoration problems, they collectively reflect a broader trend toward task-aligned learning objectives and efficiency-aware model design. Recent research [40, 41] suggests that architectural complexity is not always necessary, especially if the learning objective is well-aligned with the task.

Our framework, SMILE, demonstrates that principled functional constraints can replace architectural complexity in lightweight LLIE.

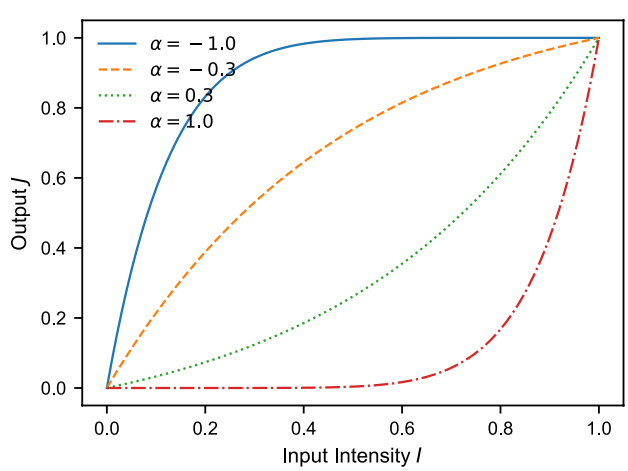

Fig. 1 Graphical representation of the Zero-DCE function for different values of the modulation factor $\alpha \in \mathcal{A}$ at iteration $n = 3$

3 Methodology

3.1 Problem statement and theoretical formulation

In LLIE, the goal is to selectively brighten underexposed regions while preserving the natural appearance of well-exposed areas. To this end, an ideal enhancement function $f(I)$ applied to a normalized image $I \in [0, 1]$ should satisfy two key properties:

- **Preservation:** If a pixel is already well-exposed, it should remain unchanged, i.e., $f(I) = I$ for such regions.
- **Enhancement:** For underexposed pixels, the function should increase brightness, i.e., $f(I) > I$ for $I \in [0, 1)$.

These requirements imply that $f(I)$ must be *monotonically increasing*, ensuring that darker pixels are mapped to brighter values while preventing reversals or reductions in well-lit areas. Formally:

$$\forall I_1, I_2 \in [0, 1], \quad I_1 < I_2 \Rightarrow f(I_1) \leq f(I_2). \tag{1}$$

However, most existing LLIE models use unconstrained modulation functions that violate this principle. A representative example is the pixel-wise update rule from Zero-DCE [18]:

$$I^{(n+1)} = I^{(n)} + A \cdot \left((I^{(n)})^2 - I^{(n)} \right), \tag{2}$$

where the modulation factor $A \in [-1, 1]$ is predicted by a neural network. While this formulation enables enhancement through iterative refinement, it suffers from directional ambiguity as illustrated in Fig. 1. When $A > 0$, the mapping becomes *monotonically decreasing*, which leads to brightness suppression rather than enhancement.

One way to mitigate this issue is to restrict A to the range $[-1, 0]$, ensuring increasing behavior. This can be achieved by applying a sigmoid activation followed by negation, i.e., $A = -\text{sigmoid}(\cdot)$. However, this workaround introduces two key limitations:

1. **Loss of Identity Mapping:** Since sigmoid never outputs zero, A cannot equal zero, and thus the model cannot fully preserve well-exposed pixels—residual brightening occurs even when no correction is needed.
2. **Limited Correction Strength:** The bounded output of sigmoid limits the modulation magnitude, making it difficult to enhance severely dark regions in a few iterations without overexposing already corrected areas.

The core issue lies in the absence of a principled constraint that enforces monotonicity in the enhancement function. Without this, the network must compensate for inconsistent behavior in its own outputs, leading to unstable training and degraded visual quality—especially in lightweight or non-reference models.

Our approach resolves this by embedding monotonicity directly into the enhancement function, ensuring consistent, stable behavior without relying on deep or skip-connected architectures.

3.2 Network overview

As shown in Fig. 2, SMILE enhances low-light image through a compact, iterative pipeline composed of two core components: a lightweight *Deep Modulation Estimation (DME)* network and a *MIF*. Given an input image $I \in \mathbb{R}^{H \times W \times 3}$, the DME network predicts a spatially adaptive modulation map $M \in \mathbb{R}^{H \times W \times 3G}$, which is split into G groups $\{M_1, M_2, \ldots, M_G\}$. Each group modulates the image in one enhancement step. The enhancement proceeds iteratively, with each stage applying a localized transformation via MIF. To improve robustness during training, we

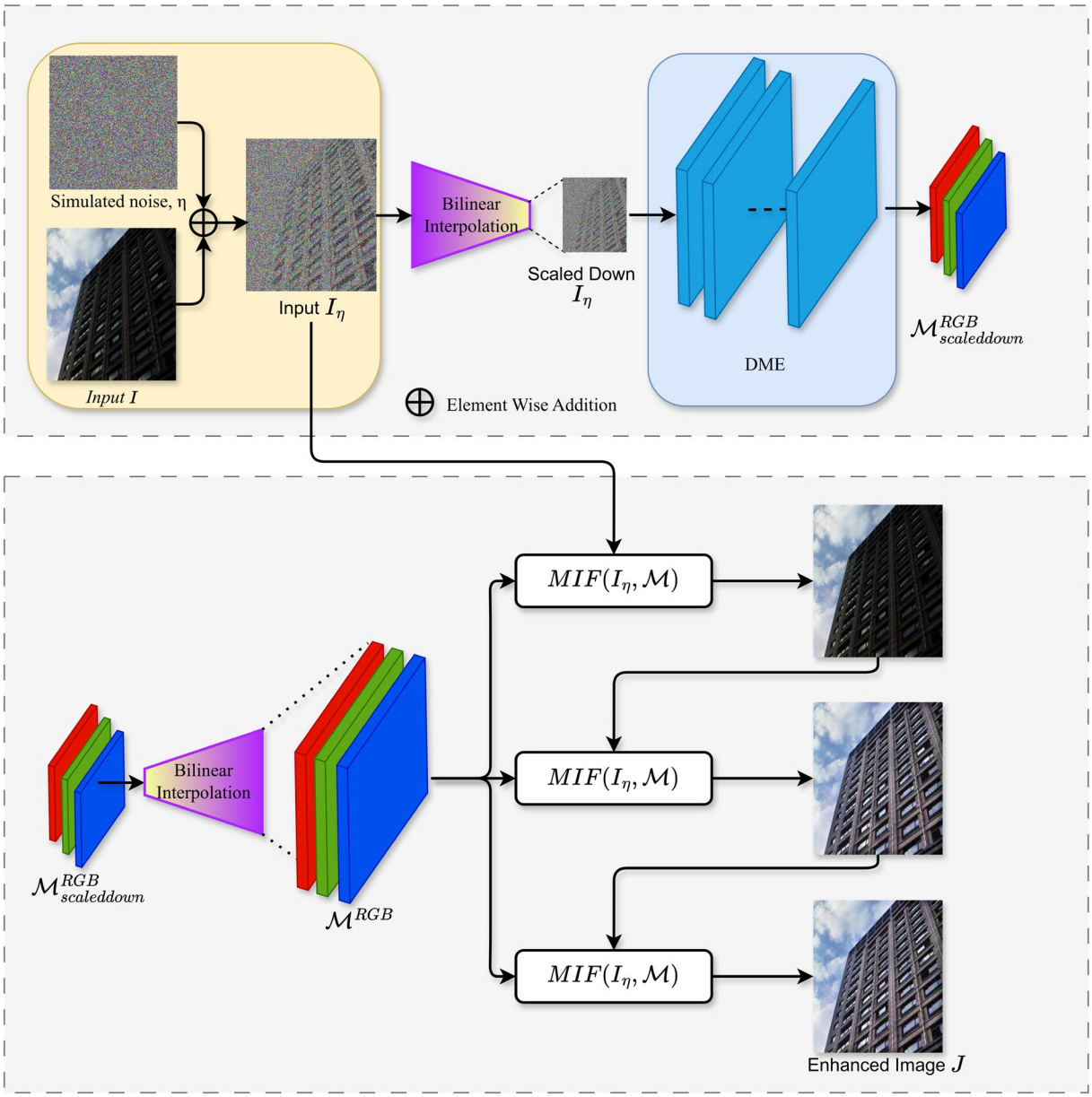

Fig. 2 Overview of SMILE. The system consists of a lightweight CNN (DME) that estimates a multi-channel modulation map $\mathcal{M}$. This modulation is applied iteratively to the input image through a nonlinear enhancement function to progressively refine brightness and contrast. Here, η denotes simulated additive noise used during training for robustness

introduce a noisy variant of the input image, denoted as $I_\eta = \text{clip}(I + \eta, 0, 1)$, where η represents simulated additive noise. In our implementation, η is sampled as zero-mean Gaussian noise with standard deviation σ (default $\sigma = \text{noise_level}$), applied independently to each pixel and color channel. This noise augmentation acts as a regularization mechanism, encouraging the model to learn stable exposure-aware modulation rather than overfitting to noise-free low-light inputs. The full procedure is detailed in Algorithm 1.

Require: Low-light image I
Ensure: Enhanced image J
1: Predict modulation map: $M \leftarrow \text{DME}(I)$
2: Split into G groups: $\{M_1, M_2, \ldots, M_G\}$
3: Initialize $J \leftarrow I$
4: **for** $g = 1$ to G **do**
5: $\quad J \leftarrow J - \frac{1-M_g}{2} \cdot (J^2 - J)$
6: **end for**
7: **return** J

Algorithm 1 Iterative Enhancement in SMILE

This iterative scheme allows SMILE to refine brightness progressively, distributing exposure correction over multiple steps. By using distinct modulation maps at each stage, the model adapts locally while avoiding abrupt global shifts.

3.3 Monotonic increasing function

MIF is central to SMILE's ability to perform stable, exposure-consistent enhancement. It modulates pixel intensities iteratively, guided by the spatially varying maps produced by DME. MIF enforces strictly increasing behavior throughout the enhancement process.

The adopted modulation function is defined as:

$$J = I - \frac{1 - M}{2} \cdot (I^2 - I), \tag{3}$$

where I and J are the input and output images, respectively, and $M \in [-1, 1]$ is the modulation map for a given iteration. Since $0 \leq 1 - M^2 \leq 1$ for all $M \in [-1, 1]$ and $I^2 - I \leq 0$ for $I \in [0, 1]$, the modulation term is non-negative. Therefore, $J \geq I$ always holds, guaranteeing monotonic enhancement and preventing brightness inversion.

This formulation introduces a symmetric scaling term, $\frac{1-M}{2}$, which ensures that the output always increases or preserves brightness, regardless of the local modulation value. As shown in Fig. 3, this design produces smooth, stable enhancement curves and eliminates the oscillatory behavior introduced by common unconstrained curve function.

Earlier variants [42], such as:

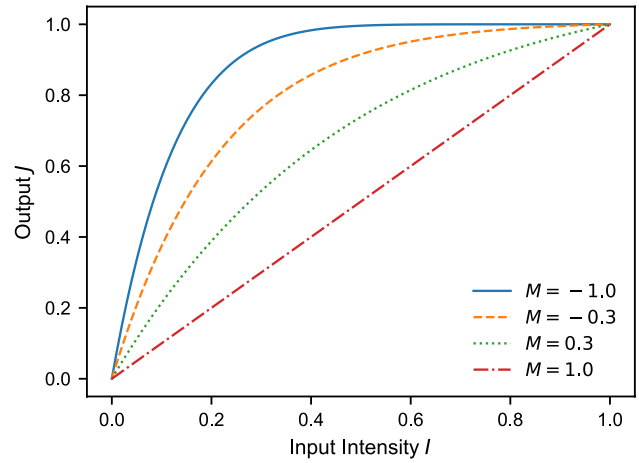

Fig. 3 Corrected enhancement curves resulting from applying Eq. 3 with $M \in [-1, 1]$ over 3 iterations. The formulation restores monotonic increasing behavior, providing consistent brightening without overshooting or oscillation

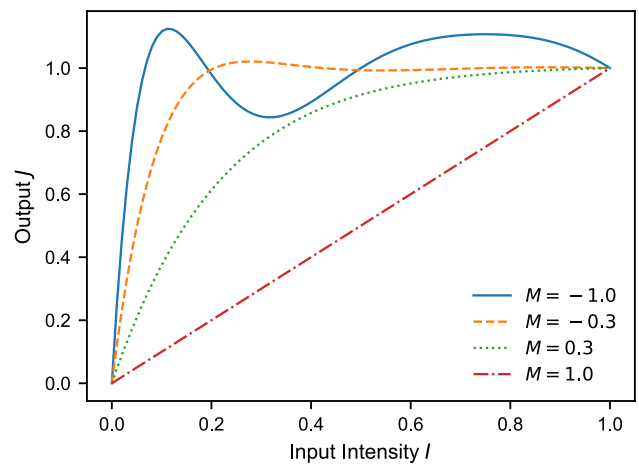

Fig. 4 Enhancement curves resulting from applying Eq. 4 with $M \in [-1, 1]$ over 3 iterations. Negative modulation values lead to oscillations and non-monotonic behavior, which can result in overexposure and structural degradation during enhancement

$$J = I - (1 - M) \cdot (I^2 - I), \tag{4}$$

demonstrated good results when $M \in [-1, 0]$. However, once deployed in a learning context where $M \in [-1, 1]$, this version becomes unstable. Figure 4 illustrates how negative values lead to non-monotonic responses that can amplify or invert brightness, especially across multiple iterations.

By contrast, the corrected formulation in SMILE guarantees monotonicity for all values of $M \in [-1, 1]$, enabling consistent enhancement across diverse luminance conditions. This removes the burden from the network to learn stability, allowing DME to focus solely on estimating how much correction is needed per region. This property is especially important in lightweight architectures, where any functional inconsistency is harder to absorb.

3.4 Deep modulation estimation network

The DME network is responsible for predicting the modulation maps used by MIF during iterative enhancement. To ensure the entire pipeline remains lightweight and stable, DME is deliberately designed as a shallow, fully convolutional network without skip connections or deep feature hierarchies. As shown in Fig. 5, DME consists of five convolutional layers with ReLU activations, followed by a final layer with a $\tanh$ activation to produce pixel-wise modulation values in the range $[-1, 1]$.

While skip connections are common in image restoration tasks to aid gradient flow and preserve high-frequency features [43–45], they also introduce architectural overhead and potentially mask weaknesses in the underlying formulation. In the context of SMILE, where functional stability is explicitly enforced through MIF and guided by UCL and EPL, we show that such architectural complexity is unnecessary. Instead, we adopt a minimal design to highlight the effectiveness of the proposed learning scheme in isolation.

Empirical results (Sect. 4) confirm that DME performs robustly without skip connections, even in resource-constrained settings.

3.5 Exposure-aware loss functions

To guide enhancement in a spatially meaningful manner, SMILE introduces two exposure-aware losses—UCL and EPL—that adapt supervision to local luminance conditions. These losses are fully differentiable and do not rely on paired training data. Instead of applying uniform penalties across all pixels, they enforce targeted behavior: correcting underexposed regions while preserving well-lit ones.

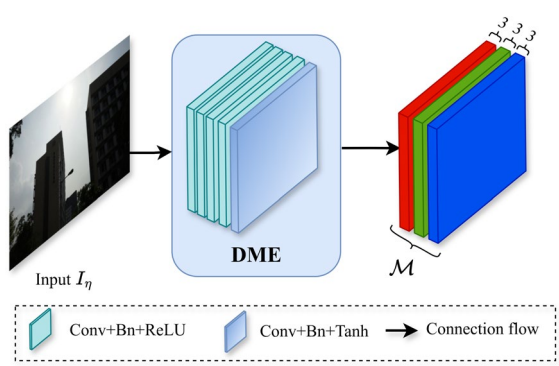

Fig. 5 Overview of the DME network used to estimate the modulation maps $\{M^g\}$. The network consists of five convolutional layers with ReLU activations and a $\tanh$ output to generate spatially adaptive pixel-wise modulation parameters in $[-1, 1]$

3.5.1 Underexposure correction loss (UCL)

UCL is designed to encourage enhancement only in regions that are genuinely dark. It operates on non-overlapping patches of the input and enhanced images, evaluating the mean luminance μ_p of each patch p. A soft mask $\sigma(k(t - \mu_p^I))$ is used to determine whether a patch is considered underexposed, where t is the luminance threshold and k controls the mask's sharpness.

The loss penalizes insufficient brightening by measuring how far the enhanced luminance μ_p^J remains below the target:

$$\mathcal{L}_{\text{UCL}} = \frac{1}{P} \sum_{p=1}^{P} \sigma(k(t - \mu_p^I)) \cdot \max(0, t - \mu_p^J)^2. \tag{5}$$

This soft formulation allows the model to enhance only where necessary, avoiding global brightening. Once the patch exceeds the target luminance, the loss automatically deactivates, promoting stable convergence and preventing overexposure.

3.5.2 Exposure preservation loss (EPL)

Complementing UCL, EPL prevents unnecessary modification in already well-exposed regions. It also uses patch-wise luminance analysis, but activates when the input brightness is above the threshold. The goal is to ensure perceptual fidelity by anchoring the output to the input where no enhancement is needed.

EPL is defined as:

$$\mathcal{L}_{\text{EPL}} = \frac{1}{P} \sum_{p=1}^{P} \sigma(k(\mu_p^I - t)) \cdot (\mu_p^J - \mu_p^I)^2. \tag{6}$$

By applying a smooth penalty on deviations from the input, EPL ensures structural consistency and avoids the introduction of artifacts in bright areas.

Together, UCL and EPL form a complementary supervision mechanism that aligns directly with the goals of LLIE. UCL focuses the network's capacity on improving visibility in dark regions, while EPL constrains it from over-adjusting already acceptable areas. This balance reduces overfitting, accelerates convergence, and eliminates the need for complex architectural mechanisms to regulate enhancement behavior.

3.6 Complementary losses and training objective

While UCL and EPL provide the primary supervision signal for exposure-aware enhancement, SMILE incorporates three complementary losses to further refine spatial coherence, edge fidelity, and color balance. These additional objectives address common perceptual and structural challenges in LLIE outputs.

3.6.1 Spatial consistency loss

To preserve local gradients and texture structures between the input and enhanced images, a spatial consistency term encourages relative intensity differences to remain locally coherent. This is defined over directional pixel pairs:

$$\mathcal{L}_{\text{spa}} = \frac{1}{N \cdot |D|} \sum_{i,j} \sum_{d \in D} \left\| (J_{i,j} - J_{i+d}) - (I_{i,j} - I_{i+d}) \right\|^2, \tag{7}$$

where (i, j) denote the indexes spatial pixel coordinates, D is a set of directional offsets (e.g., horizontal, vertical, diagonal), and N is the number of valid pixels. This loss helps suppress unnatural edge transitions and reinforces structural continuity.

3.6.2 Total variation loss

To reduce noise and promote smoothness in flat regions, total variation regularization is applied to the enhanced image:

$$\mathcal{L}_{\text{TV}} = \sum_{i,j} \left[(J_{i,j+1} - J_{i,j})^2 + (J_{i+1,j} - J_{i,j})^2 \right]. \tag{8}$$

This term penalizes abrupt intensity fluctuations, particularly effective in denoising low-light textures without introducing blur.

3.6.3 Color constancy loss

To enforce natural color balance, especially in scenes where LLIE can over-amplify certain channels, a color constancy term minimizes the difference between average RGB channel intensities:

$$\mathcal{L}_{\text{col}} = \sqrt{\sum_{(a,b) \in P} (\mu_a - \mu_b)^4}, \tag{9}$$

where μ_a, μ_b are the global mean intensities of color channels, and $P = \{(r, g), (r, b), (g, b)\}$ is the set of channel pairs. This formulation prevents color bias and preserves neutral tones.

3.6.4 Full training objective

The overall training loss is a weighted combination of all components:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{ucl}} \mathcal{L}_{\text{UCL}} + \lambda_{\text{epl}} \mathcal{L}_{\text{EPL}} + \lambda_{\text{spa}} \mathcal{L}_{\text{spa}} + \lambda_{\text{TV}} \mathcal{L}_{\text{TV}} + \lambda_{\text{col}} \mathcal{L}_{\text{col}}. \tag{10}$$

Each coefficient λ modulates the influence of its corresponding term. In practice, these weights are selected based on preliminary convergence behavior and perceptual outcomes, as detailed in Sect. 4.

Joint optimization of these objectives enables SMILE to produce high-quality, perceptually consistent results across a range of illumination conditions.

4 Experiments and results

4.1 Datasets and setup

We train SMILE using a diverse dataset [19] of 2,002 unpaired low-light images covering a broad range of luminance conditions. For evaluation, we use the LOL [46], LIME [47], VV ,[1] and DICM [48] datasets, which span both synthetic and real-world LLIE scenarios.

All experiments are implemented in PyTorch and executed on an NVIDIA RTX 3090 GPU. The model is trained for 100 epochs using AdamW [49] with a learning rate of 1×10^{-4}, weight decay of 0.01, and batch size 32. Patch-wise UCL and EPL are computed using non-overlapping 64×64 regions, with a luminance threshold $t = 0.7$ and sharpness factor $k = 50$. Loss weights are set as: $\lambda_{\text{ucl}} = 100$, $\lambda_{\text{epl}} = 100$, $\lambda_{\text{spa}} = 500$, $\lambda_{\text{TV}} = 20$, $\lambda_{\text{col}} = 10$.

4.2 Quantitative evaluation

We assess performance using both reference-based metrics—PSNR [50], SSIM [51], MSE, FSIM [52], and LPIPS [53] measured using VGG16 [54]—on datasets with paired ground truth (e.g., LOL), and no-reference metrics—NIQE [55], BRISQUE [11], and PIQE [56] on real-world datasets without ground truth (e.g., LIME and VV). SMILE is compared against strong baselines including RUAS [57], SCI [14], EnlightenGAN [58], Zero-DCE++ [19], DUAL [59], LIME [47], Kind++ [60], and UTVNet [61].

[1] https://sites.google.com/site/vonikakis/datasets.

Table 1 Comparison of various enhancement methods on the LIME, VV, and LOL datasets

Dataset	Metric	SMILE	RUAS	EnGAN	SCI	Z-DCE	DUAL	LIME	Kind	UTVNet
LIME	NIQE↓	**3.74**	4.67	3.83	3.82	3.76	3.82	3.84	3.77	3.77
	BRIS.↓	**15.68**	31.14	17.49	15.90	16.09	21.55	17.13	22.79	17.78
	PIQE↓	**30.97**	47.81	33.64	33.52	33.45	30.98	36.37	43.01	34.25
VV	NIQE↓	2.85	4.93	**2.40**	3.15	2.67	3.67	2.91	2.47	3.03
	BRIS.↓	**8.39**	30.94	10.03	8.41	9.43	13.58	7.87	8.58	10.28
	PIQE↓	**23.22**	37.16	26.80	27.26	25.65	35.43	35.30	33.38	31.99
LOL	PSNR↑	**18.52**	12.79	17.80	16.53	17.01	15.95	18.49	17.80	18.50
	SSIM↑	**0.80**	0.65	0.69	0.77	0.68	0.68	0.74	0.73	0.79
	MSE↓	**171**	176	182	188	178	181	183	174	175
	LPIPS↓	**0.32**	0.34	0.48	0.33	0.37	0.43	0.38	0.42	0.33
	FSIM↑	**0.89**	0.87	0.79	**0.89**	0.81	0.79	0.81	0.78	0.88

Due to space constraints, some names have been abbreviated: EnlightenGAN → EnGAN, Zero-DCE++ → Z-DCE, Kind++ → Kind. The BRISQUE metric is abbreviated as BRIS. The **best** and second-best results are highlighted in bold and underlined, respectively. Arrows (↑/↓) indicate whether higher or lower values are preferred

Table 2 Quantitative comparisons in terms of computational complexity

Method	Param	Mem. (MB)↓	Inf. (s)↓
Kind++	8.27M	33.1	65.29
SCI	258	8.44	2.85
Zero-DCE++	10.56 k	153.66	1.61
RUAS	3 k	164.66	1.73
UTVNet	7.75M	333.4	3.73
EnlightenGAN	8.6M	32.9	0.60
LIME	N/A	376.46	362.74
DUAL	N/A	409.89	612.03
SMILE (Ours)	3.29k	**2.93**	**0.37**

It includes the number of trainable parameters (Param.), memory usage (Mem.), and inference time (Inf.). The **best** and second-best results are highlighted in bold and underlined, respectively

These baselines represent a range of LLIE strategies: curve-based, GAN-driven, and Retinex-inspired.

Perceptual quality results in Table 1 demonstrate that SMILE achieves superior scores across all datasets and metrics. On LOL, it ranks first in PSNR, SSIM, LPIPS, and FSIM. On LIME and VV, SMILE consistently outperforms heavier models in NIQE, BRISQUE, and PIQE. These results validate that functional alignment through UCL, EPL, and MIF can replace architectural complexity.

Computational efficiency results in Table 2 show that SMILE achieves the lowest inference time and memory footprint among all models. This reflects the compactness and convergence stability enabled by MIF and exposure-aware losses.

4.3 Qualitative evaluation

Figure 6 presents visual comparisons on the VV dataset. Competing methods suffer from typical LLIE artifacts:

- SCI, UTVNet, and Zero-DCE++ produce minor overexposed or color-shifted outputs,
- EnlightenGAN and DUAL introduce structural distortions or haloing,
- RUAS causes excessive brightness in already lit regions.

In contrast, SMILE preserves global consistency while selectively enhancing dark regions. Thanks to UCL and EPL, well-lit areas remain stable, while MIF ensures that enhancement remains smooth and monotonic.

Zoomed-in insets confirm that SMILE retains fine details without introducing noise, halos, or unnatural color balance—despite its minimalist architecture.

4.4 Ablation study

To isolate the contributions of each component in SMILE, we conduct a detailed ablation study using the DICM dataset and no-reference metrics. Results are reported in Table 3.

4.4.1 Effect of exposure-aware losses

The effectiveness of each loss component is evaluated in Table 3, rows 1–3 under the Loss section. Removing UCL or EPL significantly degrades performance. Without UCL, dark regions are insufficiently corrected; without EPL, over-brightening occurs in already exposed areas. Removing both prevents stable convergence entirely. These effects are illustrated visually in Fig. 7.

4.4.2 Sensitivity analysis of luminance threshold

The luminance threshold t plays a key role in both the UCL and the EPL by determining which image regions are treated

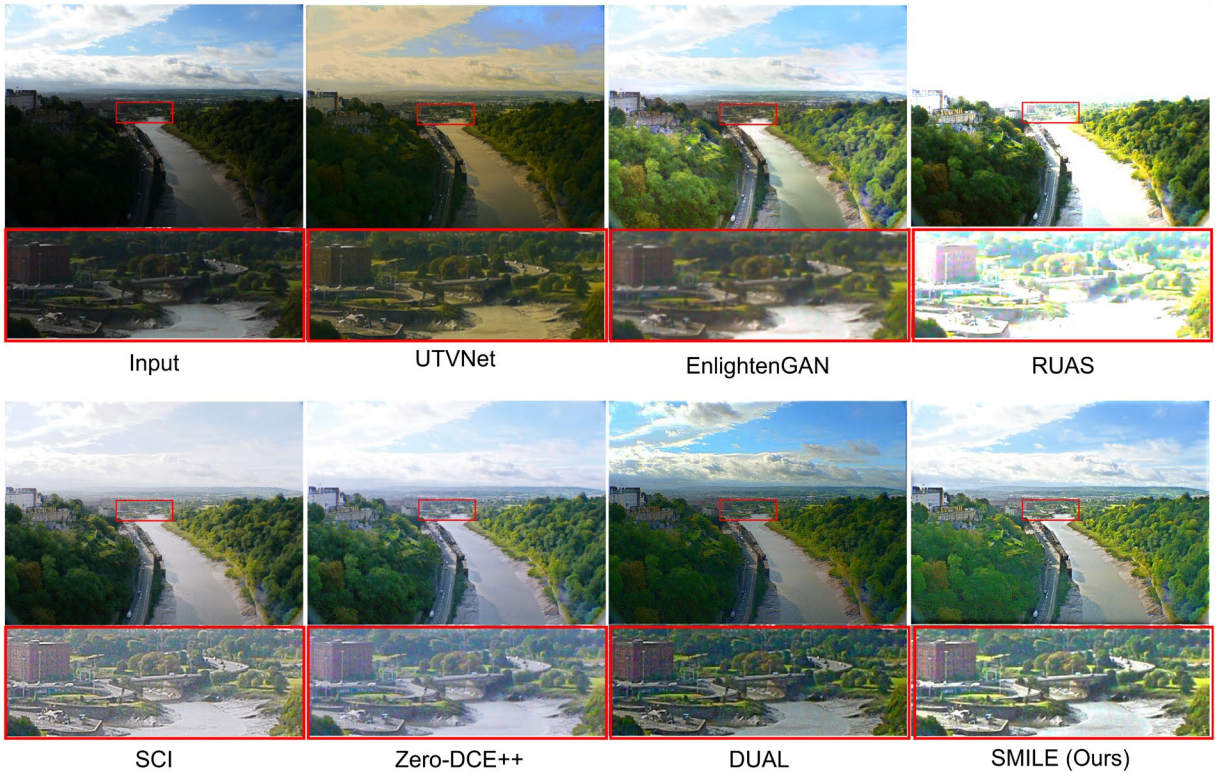

Fig. 6 Visual comparison using the VV dataset. Outputs from SCI, Zero-DCE++, UTVNet, and RUAS show noticeable color distortion or overexposure in the sky region, while EnlightenGAN and DUAL introduce major artifacts and structural inconsistencies. The zoomed-in sections, enclosed in red boxes, highlight fine-grained differences in enhancement quality. KIND++ and LIME are omitted due to space constraints but are included in the quantitative evaluation

Table 3 Ablation study results grouped by design factor

Type	Variant	NIQE↓	BRIS.↓	PIQE↓
Loss	w/o UCL	5.10	30.73	45.55
	w/o EPL	4.32	26.73	43.23
	w/o EAL	5.18	30.80	42.41
Iteration	SMILE i=2	3.87	20.98	38.77
	SMILE i=4	3.82	16.02	38.59
	SMILE1 i=3	3.83	<u>15.26</u>	34.55
Architecture	ResDME	3.84	16.04	34.26
	SkipDME	3.81	15.37	<u>34.12</u>
Modulation	SCF	**3.75**	15.44	35.11
	SCFBase	3.85	21.51	37.43
SMILE (Baseline)		<u>3.78</u>	**15.11**	**33.63**

Due to space constraints, BRISQUE have been abbreviated as BRIS. The best and second-best results are highlighted in bold and underlined, respectively

as underexposed. In all experiments, we set $t = 0.7$ as a default value. To evaluate the sensitivity of the proposed method to this parameter, we vary t from 0.5 to 0.9 while keeping all other settings unchanged.[2]

Quantitative results on the LOL dataset are reported in Table 4. When t is set to values close to 0.7 (e.g., $t = 0.6$),

[2] The best and second-best results are highlighted in bold and underlined, respectively

Fig. 7 Visual comparison showing the impact of removing UCL and EPL. The zoomed-in sections, enclosed in red boxes, highlight fine-grained differences in enhancement quality

Table 4 Sensitivity analysis of the luminance threshold t on the LOL dataset

threshold t	0.5	0.6	**0.7**	0.8	0.9
PSNR ↑	17.94	18.41	**18.52**	18.33	17.88
SSIM ↑	0.77	0.79	**0.80**	0.79	0.76

The best and second-best results are highlighted in bold and underlined, respectively

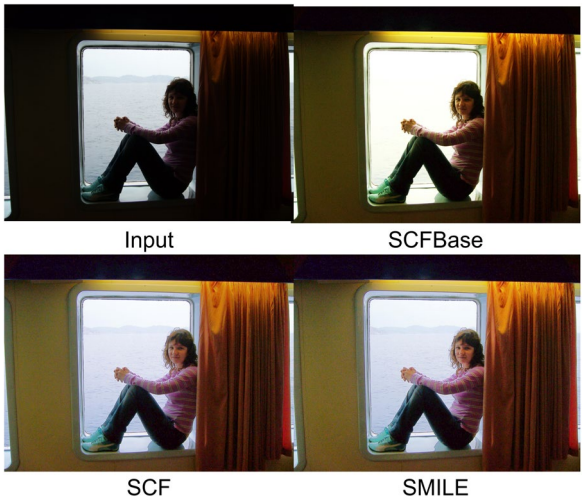

Fig. 8 Visual comparison of modulation functions. Left to right, and top to bottom: Input, SCFBase, SCF, and SMILE

performance remains stable, indicating that SMILE is not highly sensitive to small variations around the default threshold. Lower values ($t \leq 0.5$) lead to under-enhancement due to insufficient activation of exposure-aware constraints, while higher values ($t \geq 0.8$) introduce mild overexposure by incorrectly treating well-exposed regions as underexposed. These results justify the choice of $t = 0.7$ as a balanced operating point.

4.4.3 Effect of MIF

Replacing the MIF with an unconstrained formulation leads to performance degradation, as shown in the Modulation section of Table 3. The SCF variant, which retains UCL and EPL but uses a non-monotonic formulation, performs worse across all metrics. The SCFBase variant, which also removes the exposure-aware losses, fails to converge, producing degraded outputs. These differences are shown in Fig. 8. These results underscore the importance of aligning the model's optimization direction with the task-specific goal of LLIE. By combining MIF with exposure-aware supervision, SMILE encourages enhancement behavior that is spatially targeted and luminance-aware, enabling effective learning even in compact architectures.

Fig. 9 Visual comparison of iteration strategies. The zoomed-in sections, enclosed in red boxes, highlight fine-grained differences in enhancement quality

4.4.4 Effect of iteration strategy

We compare three enhancement schedules: two iterations, four iterations, and three iterations with shared modulation. Three distinct steps offer the best trade-off between refinement and stability. Table 3 and Fig. 9 support this conclusion. Fewer iterations under-enhance, while more introduce softness or redundancy.

4.4.5 Effect of network architecture

As shown in the Architecture section of Table 3, adding skip or residual connections to DME yields negligible improvements (or slight regressions). This supports our core claim: when supervision is exposure-aligned and the modulation function is stable, architectural complexity is not required for strong LLIE performance.

5 Conclusion

This work presented **SMILE**, a lightweight and non-reference framework for LLIE that achieves SOTA performance with only 3.3k parameters. Unlike prior methods that rely on architectural complexity to compensate for task misalignment, SMILE is built upon principled design choices: MIF ensures stable, monotonic enhancement, while UCL and EPL provide exposure-aware supervision at the patch level.

Through extensive experiments and ablations, we demonstrated that SMILE delivers strong perceptual quality, fast inference, and high convergence stability—despite operating without paired data or skip connections. These

results confirm that targeted loss functions and functional constraints can outperform purely architectural strategies in both effectiveness and efficiency.

Beyond LLIE, the underlying approach of guiding enhancement based on localized input characteristics offers a general direction for future research in lightweight, task-aligned vision systems. Although SMILE is designed for low-light enhancement, the proposed exposure-aware perception mechanism may be extended to other image restoration tasks such as dehazing or low-level super-resolution, where spatially adaptive correction is also critical. We leave such extensions as future work.

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Data Availability The datasets used for training and testing in this study are publicly available. The training dataset can be accessed at the following link: Training dataset (https://drive.google.com/file/d/1GAB3uGsmAyLgtDBDONbil08vVu5wJcG3/view), LOL test set (https://www.kaggle.com/datasets/soumikrakshit/lol-dataset), LIME, DICM, datasets (https://drive.google.com/drive/folders/0B_FjaR958nw_djVQanJqeEhUM1k), LIME, DICM, NPE, MEF datasets (https://drive.google.com/drive/folders/0B_FjaR958nw_djVQanJqeEhUM1k), VV dataset (https://drive.google.com/file/d/1RL7sf0vvfE7UDSQUH7atHHkibAUjprtn/view?usp=sharing). All datasets are publicly accessible, and their respective sources have been cited accordingly.

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